

**Material Safety Data Sheet****KODIAK® CONCENTRATE BIOLOGICAL FUNGICIDE**

MSDS Number: 102000012740

MSDS Version 2.0

Revision Date: 07/15/2008

SECTION 1. CHEMICAL PRODUCT AND COMPANY INFORMATION

Product Name KODIAK® CONCENTRATE BIOLOGICAL FUNGICIDE
MSDS Number 102000012740
EPA Registration No. 264-951

Bayer CropScience
2 T.W. Alexander Drive
Research Triangle PK, NC 27709
USA

For MEDICAL, TRANSPORTATION or other EMERGENCY call: 1-800-334-7577 (24 hours/day)
For Product Information call: 1-866-99BAYER (1-866-992-2937)

SECTION 2. COMPOSITION/INFORMATION ON INGREDIENTS

<u>Hazardous Component Name</u>	<u>CAS-No.</u>	<u>Average % by Weight</u>
Bacillus subtilis GB03 (contains 5.5 x 10 exp10 CFU per gram)	68038-70-0	2.75

SECTION 3. HAZARDS IDENTIFICATION

NOTE: Please refer to Section 11 for detailed toxicological information.

Emergency Overview Caution! Moderate eye irritation. Causes skin irritation. Prolonged or frequently repeated skin contact may cause allergic reactions in some individuals. May cause sensitization by inhalation and skin contact. Avoid contact with skin, eyes and clothing. Avoid breathing dust.

Physical State powder
Odor musty
Appearance off-white to beige
Routes of Exposure Inhalation, Eye contact, Skin contact

SECTION 4. FIRST AID MEASURES

General When possible, have the product container or label with you when calling a poison control center or doctor or going for treatment.

Eye Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. Call a physician or poison control center immediately.

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Skin	Take off contaminated clothing and shoes immediately. Wash off immediately with plenty of water for at least 15 minutes. Call a physician or poison control center immediately.
Ingestion	Call a physician or poison control center immediately. Rinse out mouth and give water in small sips to drink. DO NOT induce vomiting unless directed to do so by a physician or poison control center. Never give anything by mouth to an unconscious person. Do not leave victim unattended.
Inhalation	Move to fresh air. If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a physician or poison control center immediately.
Notes to Physician Treatment	There is no specific antidote. Appropriate supportive and symptomatic treatment as indicated by the patient's condition is recommended.

SECTION 5. FIRE FIGHTING MEASURES

Flash point	not applicable
Suitable Extinguishing Media	Foam, Carbon dioxide (CO ₂), Dry chemical, Water spray
Fire Fighting Instructions	Keep out of smoke. Fight fire from upwind position. Cool closed containers/ tanks exposed to fire with water spray. Do not allow run-off from fire fighting to enter drains or water courses. Firefighters should wear NIOSH approved self-contained breathing apparatus and full protective clothing.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal Precautions	Keep unauthorized people away. Isolate hazard area. Avoid contact with spilled product or contaminated surfaces.
Methods for Cleaning Up	Sweep up or vacuum up spillage and collect in suitable container for disposal. Clean contaminated floors and objects thoroughly, observing environmental regulations.
Additional Advice	Use personal protective equipment. Do not allow to enter soil, waterways or waste water canal.

SECTION 7. HANDLING AND STORAGE

Handling Procedures	Use only in area provided with appropriate exhaust ventilation.
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Storing Procedures	Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from food, drink and animal feedingstuffs.
Work/Hygienic Procedures	Wash hands thoroughly with soap and water after handling and before eating, drinking, chewing gum, using tobacco, using the toilet or applying cosmetics. Remove Personal Protective Equipment (PPE) immediately after handling this product. Before removing gloves clean them with soap and water. Remove soiled clothing immediately and clean thoroughly before using again. Wash thoroughly and put on clean clothing.
Min/Max Storage Temperatures	Recommended minimum transport/storage temperature: 0 °C / 32 °F Recommended maximum transport/storage temperature: 43 °C / 109 °F

SECTION 8. EXPOSURE CONTROLS / PERSONAL PROTECTION

General Protection	Follow all label instructions. Train employees in safe use of the product. Follow manufacturer's instructions for cleaning/maintaining PPE. If no such instructions for washables, use detergent and warm/tepid water. Keep and wash PPE separately from other laundry.			
Eye/Face Protection	Tightly fitting safety goggles			
Hand Protection	Chemical resistant nitrile rubber gloves			
Body Protection	Wear long-sleeved shirt and long pants and shoes plus socks.			
Respiratory Protection	When respirators are required, select NIOSH approved equipment based on actual or potential airborne concentrations and in accordance with the appropriate regulatory standards and/or industry recommendations.			
Exposure Limits				
Kaolin	1332-58-7	ACGIH	TWA	2 mg/m ³
		NIOSH	REL	5 mg/m ³
		NIOSH	REL	10 mg/m ³
		US CA OEL	TWA PEL	2 mg/m ³
		TX ESL	ST ESL	20 ug/m ³
		TX ESL	AN ESL	2 ug/m ³

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MSDS Version 2.0**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

Appearance	off-white to beige
Physical State	powder
Odor	musty
pH	4.8 (1 %) suspension in water
Bulk Density	17 - 26 lb/ft ³
Water solubility	suspensive

SECTION 10. STABILITY AND REACTIVITY

Incompatibility	Acids Bases Oxidizing agents
Hazardous Reactions	No dangerous reaction known under conditions of normal use. Stable under recommended storage conditions.

SECTION 11. TOXICOLOGICAL INFORMATION

Acute Oral Toxicity	Not toxic or pathogenic to rats exposed to 10 exp 8 CFU (colony forming units)
Acute Dermal Toxicity	rabbit: LD50: > 2,000 mg/kg
Acute Inhalation Toxicity	Not toxic or pathogenic to rats exposed to 10 exp 8 CFU (colony forming units)
Skin Irritation	rabbit: Moderate skin irritation.
Eye Irritation	rabbit: Moderate eye irritation.

Assessment Carcinogenicity

ACGIH
None.
NTP
None.
IARC
None.
OSHA
None.

SECTION 12. ECOLOGICAL INFORMATION

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**Environmental
Precautions**

Do not apply directly to water, to areas where surface water is present or to intertidal areas below the mean high water mark. Do not contaminate surface or ground water by cleaning equipment or disposal of wastes, including equipment wash water.

SECTION 13. DISPOSAL CONSIDERATIONS**General Disposal
Guidance**

Follow container label instructions for disposal of wastes generated during use in compliance with the product label.

Container Disposal

Do not re-use empty containers. Follow advice on product label and/or leaflet.

SECTION 14. TRANSPORT INFORMATION**TRANSPORTATION CLASSIFICATION:**

Not regulated for all modes of transportation

FREIGHT CLASSIFICATION:

Bacteria, Seed or Soil inoculant

SECTION 15. REGULATORY INFORMATION

EPA Registration No. 264-951

US Federal Regulations**TSCA list**

Bacillus subtilis GB03

68038-70-0

US. Toxic Substances Control Act (TSCA) Section 12(b) Export Notification (40 CFR 707, Subpt D)

None.

SARA Title III - Section 302 - Notification and Information

None.

SARA Title III - Section 313 - Toxic Chemical Release Reporting

None.

US States Regulatory Reporting**CA Prop65**

This product does not contain any substances known to the State of California to cause cancer.

This product does not contain any substances known to the State of California to cause reproductive harm.

US State Right-To-Know Ingredients

None.

Canadian Regulations**Canadian Domestic Substance List**



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Environmental

CERCLA

None.

Clean Water Section 307 Priority Pollutants

None.

Safe Drinking Water Act Maximum Contaminant Levels

None.

SECTION 16. OTHER INFORMATION

NFPA 704 (National Fire Protection Association):

Health - 2 Flammability - 0 Reactivity - 0 Others - none

0 = minimal hazard, 1 = slight hazard, 2 = moderate hazard, 3 = severe hazard, 4 = extreme hazard

Reason to Revise: Updated due to new system and numbering scheme. Updated for general editorial purposes.

Revision Date: 07/15/2008

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